

Sleep Duration and Memory Retention in Adult Learners

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Abstract

This controlled study examined the relationship between sleep duration and next-day vocabulary recall in adult learners. Eighty adult learners were randomly assigned to a 5-hour sleep opportunity or an 8-hour sleep opportunity after studying a list of 120 foreign-language vocabulary pairs.

Methods

Participants completed a delayed recall test the next morning. The 8-hour sleep group had a mean vocabulary recall rate of 86 percent. The 5-hour sleep group had a mean vocabulary recall rate of 71 percent. The difference between groups was 15 percentage points. Mean reaction time during the recall task was 420 milliseconds in the 8-hour sleep group and 460 milliseconds in the 5-hour sleep group.

Conclusion

Longer sleep duration was associated with better next-day vocabulary retention in this sample. The study did not test caffeine intake, mindfulness training, cortisol, or long-term academic grades.